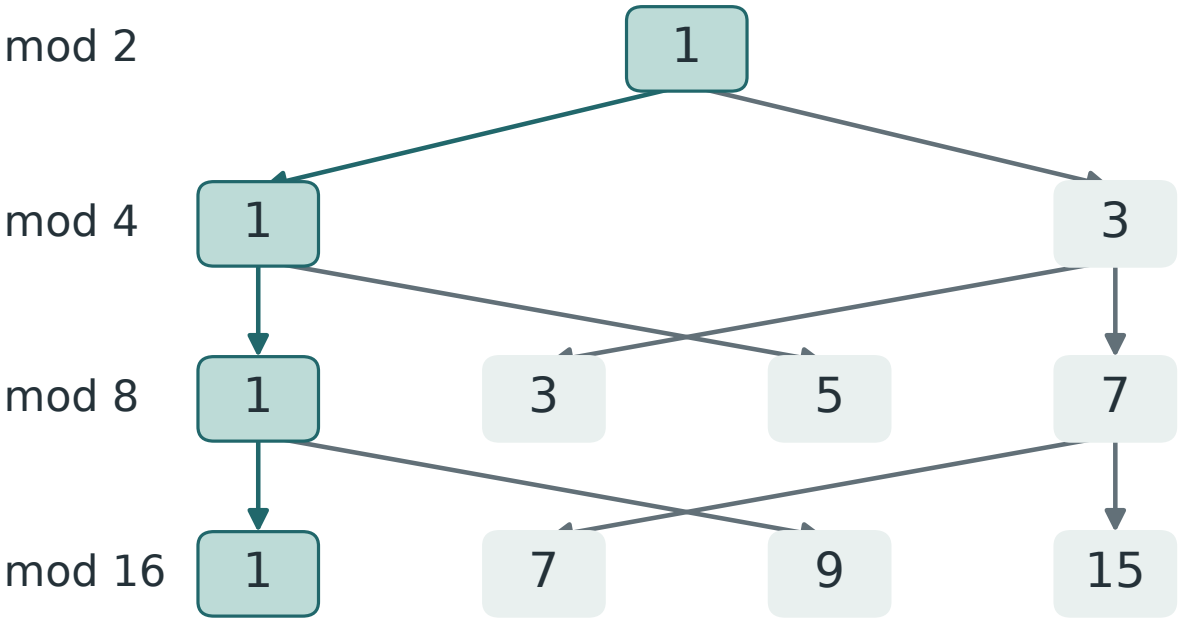
Every root of $x^2 = 1$ modulo 2, 4, 8, 16



3 mod 8 has no successor mod 16